

Balance Constants, Majority Cycles, and the Gold Partition Conjecture through Fourteen Elements

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Abstract

We determine the exact extremal balance data of all 1,338,193,159,771 unlabeled posets on fourteen elements. The least balance constant exceeding $1/3$ is $37/106$. The least over posets that are not nontrivial ordinal sums is $254/725$, attained by a ladder with broken rungs; this confirms a conjecture of Peczarski at order 14, while orders 12 and 13 reproduce De Loof, De Baets, and De Meyer. No balance constant lies in the gap above $1/3$ that Peczarski conjectures to be empty. Exactly 128 classes attain $1/3$, and every one is an ordinal sum of singletons and copies of the three-element poset with one relation, a family whose non-chain members are counted by $a(n) - 1$, where $a(n) = a(n - 1) + a(n - 3)$. In the linear-extension-majority digraph the longest simple cycle has length 8, against 7 at order 13, and exactly 30 classes attain it; of the thirteen such classes whose witnesses the census retains, nine have a cycle spectrum containing no odd cycle at all. A second exhaustive pass over the same classes verifies Peczarski's Gold Partition Conjecture through fourteen elements, extending his order-11 frontier and implying in particular that the $1/3$ – $2/3$ Conjecture holds through order 14. All arithmetic is exact and every extremal witness is recomputed by an independent program.

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1 Introduction

Let P be a finite poset on n elements and write $e(P)$ for its number of linear extensions. For $x, y \in P$, let $P + xy$ denote the poset obtained by adjoining $x < y$ and taking the transitive closure; if $y < x$ already, set $e(P + xy) = 0$. Thus

$$\Pr[x \prec y] = \frac{e(P + xy)}{e(P)}$$

is the probability that x precedes y in a uniformly chosen linear extension. For a non-chain P , its *balance constant* is

$$b(P) = \max_{x \parallel y} \min\{\Pr[x \prec y], \Pr[y \prec x]\},$$

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the maximum over incomparable pairs. The 1/3–2/3 Conjecture asserts that $b(P) \geq 1/3$ for every finite non-chain. It was independently posed by Kislitsyn, Fredman, and Linial in connection with sorting under partial information [18, 12, 19]. Kahn and Saks [17] first proved a uniform positive bound, improved by Brightwell, Felsner, and Trotter [5]; Chen [8] gives width-two constructions near the conjectured gap, while Sah [26] gives the sharpest known quantitative width-two bound. Olson and Sagan [21] treat several structured families, and Aires and Kahn [3, 4] have recently obtained large-width and bounded-width results. Chan and Pak [7] survey the area.

Two exhaustive computations over the order-14 classes are reported here. The first is a census of exact balance and majority-cycle data (Sections 3–4); the second decides a stronger conjecture of Peczarski for every class (Section 5). They share the generator and the ideal-lattice machinery of Section 2 but are separate passes with separate binaries, and we keep their evidence separate.

Running both permits a comparison that neither pass records directly. The Gold Partition run counts certificates but not their posets; the census records extremal posets but not their certificates. Section 6 classifies the retained extremal witnesses: many low-tail witnesses need a triple certificate, whereas all 37 retained length-7 or length-8 witnesses have a pair certificate. Their deterministic retention rules are biased, so this comparison does not estimate class-wide conditional rates.

1.1 Extremal balance

Peczarski [25] studied which posets are worst balanced. He observed that the global minimum of b above $1/3$ is not informative on its own: forming the ordinal sum of P with a singleton leaves b unchanged, so the minimum is inherited from smaller orders rather than attained afresh. Restricting to posets that are not nontrivial ordinal sums — equivalently, not linear sums — he conjectured that the worst is always a *ladder with broken rungs*, and separately that no poset has balance constant strictly between $1/3$ and approximately 0.348843. He verified the first at orders 3, 4, 6, 9, 10, and 11 and conjectured it for every order except 5, 7, and 8, which he excludes explicitly; at $n \geq 12$ he restricted his searches to width two, width three, and the ladder classes, judging a complete pass infeasible.

Theorem 1. *The minimum balance constant over the order-14 posets that are not nontrivial ordinal sums is $254/725$, attained by the ladder $L_{14,1,9}$.*

De Loof, De Baets, and De Meyer determined the same quantity — they call such posets *worst balanced* — for every order up to 13 [10, §4.2.2], and observed that orders 12 and 13 also fall into Peczarski’s class. Theorem 1 extends that by one order; our orders 12 and 13 reproduce their result and are reported as a regression, not as a contribution.

Theorem 2. *No poset on at most 14 elements has balance constant strictly between $1/3$ and 0.348843. The least value above $1/3$ at order 14 is $37/106 = 0.349056\dots$, inherited from a ten-element poset.*

Here 0.348843 is the decimal printed in [25] for a conjectured limit, not an exact rational threshold, so Theorem 2 verifies the conjecture at these orders rather than saying anything about the limit itself. A gap of this kind was asked for by Brightwell [2] and is a theorem in width two: Sah [26] proves $b(P) \geq (-3 + 5\sqrt{17})/52 = 0.33876\dots$ for every width-two poset that is not such an ordinal sum. Theorem 2 is weaker in reach and stronger in constant — it holds over all widths but only to order 14, and it clears Peczarski’s larger threshold rather than Sah’s proved one.

Proposition 3. *Let T be the three-element poset consisting of a two-element chain and an isolated point. Every ordinal sum of singletons and copies of T , other than the chain, has balance constant exactly $1/3$, and the number of such posets on n elements is $a(n) - 1$, where*

$$a(n) = a(n-1) + a(n-3), \quad a(0) = a(1) = a(2) = 1.$$

No other poset on at most 14 elements has balance constant $1/3$.

The converse is a theorem in width two: Aigner [1] showed that the width-two posets with $b(P) = 1/3$ are exactly the ordinal sums of singletons and copies of T . Every one of the 128 order-14 equality classes has width two, so at order 14 the census does not extend Aigner’s classification within width two; what it adds is that no poset of any larger width attains $1/3$ either, which his argument does not cover.

For $n = 3, \dots, 14$ the counts are 1, 2, 3, 5, 8, 12, 18, 27, 40, 59, 87, 128.

1.2 Majority cycles

The *linear-extension-majority* (LEM) digraph $D(P)$ has vertex set P and an edge $x \rightarrow y$ whenever $\Pr[x \prec y] > 1/2$. It has no loop and no 2-cycle, since $\Pr[x \prec y] + \Pr[y \prec x] = 1$, but it need not be acyclic: a directed cycle is a Condorcet-style failure of transitivity for the majority relation. The majority relation appeared in Kisilitsyn [18]; Fishburn exhibited cycles in 1974 [13] and later studied its majority graphs [14]. Gehrlein and Fishburn made the first exhaustive search for cycles, through order 9 [15]. De Loof, De Baets, and De Meyer [10] counted the posets carrying such a cycle, by length, through order 13, where the longest has length 7 and is attained by exactly one class.

Theorem 4. *The longest simple cycle in the linear-extension-majority digraph of an order-14 poset has length 8. Exactly 30 classes carry one.*

1.3 The Gold Partition Conjecture

The Gold Partition Conjecture is stronger than the $1/3$ – $2/3$ Conjecture. It asks for two comparisons chosen in advance such that $t_0 \geq t_1 + t_2$ for every sequence of outcomes, where t_0 is the initial number of linear extensions and t_i the number remaining after the first i comparisons, with $t_2 = 1$ when the first comparison leaves a chain. Peczarski introduced it, proved that it implies the $1/3$ – $2/3$ Conjecture, and verified it through 11 elements [22, Conjecture 1, Proposition 1, and Theorem 3]; the name refers to the resulting sorting bound $C(P) \leq \log_\varphi e(P)$ with $\varphi = (1 + \sqrt{5})/2$ [22, Propositions 2–3]. We are aware of no later exhaustive extension.

Theorem 5. *Every non-chain poset on at most 14 elements satisfies the Gold Partition Conjecture.*

Corollary 6. *Every non-chain poset on at most 14 elements has balance constant at least $1/3$.*

Corollary 6 follows from Theorem 5 and Peczarski’s Proposition 1, and is confirmed independently by the census, which computes $b(P)$ exactly for every class and finds no value below $1/3$. Only its order-14 case advances the previous computational frontier: De Loof, De Baets, and De Meyer computed all mutual rank probabilities and determined the worst balanced posets through order 13 [10].

Structurally, Peczarski proved the conjecture for posets with a nontrivial automorphism, for a class containing the N -free posets, and for 6-thin posets [24, 23], and Dolores-Cuenca, Guzmán-Sáenz, and Kim showed it is preserved by lexicographic sum [11, Lemma 2.5], an argument that applies unchanged to the non-strict inequality.

1.4 What is new

Relative to the literature, the complete balance and majority-cycle census is new at order 14, while the Gold Partition verification is new at orders 12–14: Peczarski’s exhaustive frontier was order 11. The worst balanced posets through order 13 reproduce De Loof et al. [10, §4.2.2], as do the cycle counts at orders 9 through 13, and the ladder values reproduce Peczarski’s Table 1 [25]. Relative to the first arXiv version of this paper, the new material is the order-14 census and the synthesis of its data with the Gold Partition certificates. The all-pairs recurrence on the lattice of order ideals is due to De Loof, De Meyer, and De Baets [9]; modulus sharding is standard, with prior art in both enumeration papers we depend on [6, 10]. Our contribution is the new exhaustive ranges, their witnesses, the certificate-directed implementation where its effect is measured, and the reproducible verification package.

2 Machinery

2.1 Exact counting

Let $\mathcal{I}(P)$ be the lattice of order ideals of P , and let $F(I)$ be the number of linear extensions of the induced poset on I . Then

$$F(\emptyset) = 1, \quad F(I) = \sum_{x \in \max(I)} F(I \setminus \{x\}). \quad (1)$$

Assigning zero to ideals that violate an adjoined comparison gives $e(P + xy)$ and $e(P + xy + yz)$.

If $B(I)$ is the number of ways to complete a linear extension whose initial ideal is I , then one forward and one backward pass supply every pair count:

$$e(P + xy) = \sum_{\substack{I \in \mathcal{I}(P) \\ x \in I, y \notin I \\ I \cup \{y\} \in \mathcal{I}(P)}} F(I)B(I \cup \{y\}). \quad (2)$$

Each product in (2) counts a subset of the linear extensions, their sum is at most $e(P)$, and the comparisons use only small integer multiples. This ideal-lattice method for obtaining all mutual rank probabilities in one forward and one backward pass is due to De Loof, De Meyer, and De Baets [9]. Our contribution is not a new all-pairs algorithm.

The two passes use it differently, and the difference is the reason there are two. The census needs every pair probability of every class, so it evaluates (2) in full for each class surviving the ordinal-sum and duality reductions of Section 3. The Gold Partition decision needs only a certificate, so it directs the same recurrences toward the cheapest sufficient test and stops there (Section 5).

All extension counts are unsigned 64-bit integers. This is sufficient through order 14, since $e(P) \leq 14! < 2^{41}$; the formulas above use only small integer multiples of these counts. Every comparison of two probabilities is an exact cross product in unsigned 128-bit arithmetic, and every balance constant is reduced and stored as a numerator–denominator pair. No floating-point value is computed, compared, or stored anywhere in a decision path; decimals appear only in printed reports. The independent checkers use Python integers and `fractions.Fraction` for the same reason.

The posets are supplied by Brinkmann and McKay’s `genposetg`, which uses `nauty` for canonical labelling [6, 20]. Both properties studied here are invariant under isomorphism, so one representative of each unlabeled class suffices. The generator’s deterministic modulus filter divides the stream into disjoint residues: the Gold Partition run used modulus 4,096 and the invocation

```
gpc 14 o q m <residue> 4096,
```

and the census used modulus 16,384. Every shard records the parameters it was compiled with and the SHA-256 of the binary that produced it; each aggregator refuses a shard set whose parameters or binary differ, verifies that the residues are disjoint and covering, and checks the total against the published class count.

Two capacities and one search budget could in principle be exceeded in the census: the tail table, the equality table, and the per-poset cycle-search node budget. Each causes a nonzero exit, leaving the shard rejected rather than silently truncated. Nothing is capped quietly.

Among all posets attaining an extremal value the census retains the lexicographically least encoding, so the reported witness does not depend on the shard count or on scheduling. One qualification: the duality filter processes one member of each dual pair, so the witness is least among the retained representatives.

2.2 Ordinal sums

Let the connected components of the incomparability graph of P be $P_1, \dots, P_r$. Any two elements in different components are comparable and, for a fixed pair of components, all such comparisons point the same way, so $P = P_1 \oplus \dots \oplus P_r$ is the finest ordinal-sum decomposition. A linear extension of an ordinal sum is exactly a concatenation of independently chosen linear extensions of the summands, so the restriction map $\mathcal{E}(P) \rightarrow \mathcal{E}(P_i)$ is surjective with all fibres of size $e(P)/e(P_i)$. Hence probabilities within a summand are computed in the summand, and

$$b(P) = \max_i b(P_i), \tag{3}$$

the maximum over non-singleton summands. Cycles likewise cannot cross summands in D_{inc} , and in D a crossing edge always points the same way, so no cycle uses one. The census decomposes and works summand by summand, which is both a large saving and the reason the non-ordinal-sum minimum is well defined.

Identity (3) has the consequence Peczarski noted, and it is easy to overlook when reading a table of minima: $b(P \oplus \mathbf{1}) = b(P)$, so the global minimum above $1/3$ is non-increasing in n and, at these orders, is inherited from a smaller summand rather than newly attained by an order- n ordinal-indecomposable poset. The value $37/106$ comes from $L_{10,1,5}$, a *ten*-element ladder, padded with singletons. The quantity that can still move with n , and the one Theorem 1 is about, is the minimum over posets that are not nontrivial ordinal sums.

3 Balance constants

3.1 The low tail

Figure 1 shows every distinct balance constant in the window $(1/3, 9/25]$, with the number of classes attaining it. The window is compiled into the binary and comfortably contains every published record.

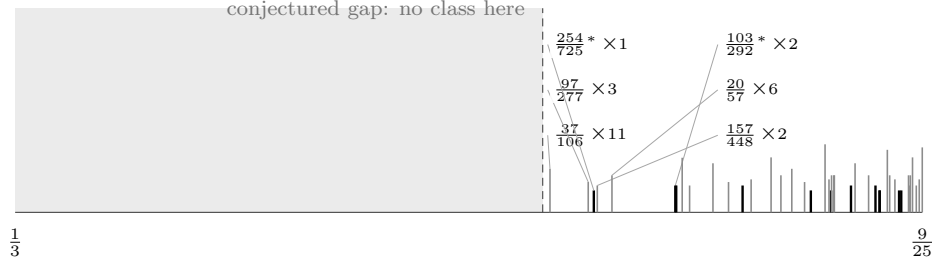


Figure 1: The low balance tail: all 42 distinct balance constants of order-14 posets in $(1/3, 9/25]$. The shaded region is the gap conjectured empty in [25]; no class lies in it. Stem height grows with the number of classes attaining the value, shown as $\times k$; values marked $*$ are attained by a poset that is not a nontrivial ordinal sum. Six values are labelled; the rest are drawn unlabelled.

At order 14 there are 42 distinct values in the window, attained by 469 classes in all. Exactly 30 are the order-13 tail inherited by ordinal-sum padding; the remaining 12 are attained by an order-14 non-sum. The values cluster immediately above the conjectured gap rather than spreading across the window: the five smallest, $37/106$, $97/277$, $254/725$, $157/448$, and $20/57$, all lie within 2×10^{-3} of each other, and the whole window holds fewer than five hundred classes out of 1.3 trillion. The 12 new values are attained by one or two non-sum classes each; their minimum is the quantity Theorem 1 concerns.

3.2 Ladders with broken rungs

For $n \geq 3$ the ladder L_n has elements $u_1 < u_2 < \dots$ and $v_1 < v_2 < \dots$ interleaved by rungs; breaking a rung deletes one cross relation. Peczarski writes $L_{n,i_1,\dots,i_k}$ for the n -element ladder with rungs $i_1, \dots, i_k$ broken [25].

We did not quote his table. For each order the program constructs every broken-rung ladder from his definition, computes its balance constant exactly, and takes the minimum; at order 14 this is an exhaustive search over all $2^{11} = 2,048$ rung subsets. The result is then compared with the census minimum over non-ordinal-sums, which is what proves Theorem 1.

n	census minimum, non-sum	worst broken-rung ladder	agree
7	14/39	$9/25 = L_{7,1,2}$	no [†]
8	16/45	$17/46 = L_{8,1,2,3}$	no [†]
9	6/17	$6/17 = L_{9,1,2,3,4}$	yes
10	37/106	$37/106 = L_{10,1,5}$	yes
11	20/57	$20/57 = L_{11,1,6}$	yes
12	97/277	$97/277 = L_{12,1,7}$	yes [‡]
13	157/448	$157/448 = L_{13,1,8}$	yes [‡]
14	254/725	$254/725 = L_{14,1,9}$	yes, new

Table 1: Peczarski’s broken-rung conjecture against the census. Each ladder value was recomputed by exhaustive search over rung subsets, not quoted. [†]: $n = 5, 7, 8$ are the exceptions [25] excludes explicitly; his conjecture is stated for every other order, and verified there at 3, 4, 6, 9, 10, and 11. [‡]: also obtained by De Loof et al. [10, §4.2.2]; our rows reproduce theirs.

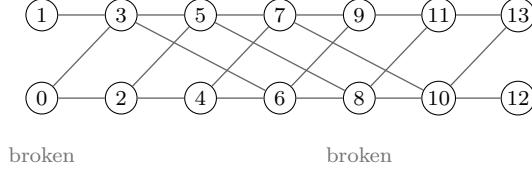


Figure 2: $L_{14,1,9}$, the worst broken-rung ladder at order 14, with $b = 254/725$ and $e = 725$. The two rungs marked *broken* are the missing cross relations.

Table 1 agrees with his Table 1 at every order where the two overlap, including the two exceptions he records.

A reader tabulating $n = 10$ through 14 will notice that the extremal ladder is $L_{n,1,n-5}$ throughout, and should know that this does not continue: it fails at $n = 9$, where four rungs break, and Peczarski's Table 1 gives $L_{15,1,5,6,10}$ and $L_{17,1,5,8,12}$.

3.3 The equality locus

Proof of Proposition 3. The poset T has $e(T) = 3$ and $b(T) = 1/3$. By (3), an ordinal sum of singletons and copies of T has balance constant $1/3$ as soon as at least one summand is a copy of T ; if all summands are singletons the sum is a chain and has no incomparable pair. Decomposition into ordinal-indecomposable summands is unique, so these posets are in bijection with the compositions of n into parts 1 and 3. Splitting on the first part gives $a(n) = a(n-1) + a(n-3)$, and discarding the all-singleton composition gives $a(n) - 1$. $\square$

The converse — that nothing else attains $1/3$ — is a computational statement, verified here at every order through 14 and, to our knowledge, not proved. The aggregator does not merely count: it recomputes each representative's balance constant and separately recognises the family structurally, by decomposing the poset and checking that every summand is a singleton or a copy of T , without computing any probability. The two recognitions must agree.

4 Majority cycles

4.1 Which majority relation

De Loof et al. define the LEM relation as the strict $1/2$ -cut of the mutual rank probability relation over *all* ordered pairs of distinct elements. A comparable pair $x < y$ has $\Pr[x \prec y] = 1$ and so contributes an edge. Writing $D_{\text{inc}}(P)$ for the sub-digraph on incomparable pairs only, $D(P)$ is $D_{\text{inc}}(P)$ together with the strict order relation of P .

One would like to work with the cheaper $D_{\text{inc}}(P)$, arguing that comparable edges cannot lie on a cycle. That is provable only for triangles.

Proposition 7. *No 3-cycle of $D(P)$ uses a comparable edge.*

Proof. Suppose $x \rightarrow y \rightarrow z \rightarrow x$ with $x < y$ in P . Then $z \rightarrow x$ and $x < y$ give $\Pr[z \prec y] \geq \Pr[z \prec x] > 1/2$, since every extension with $z \prec x$ also has $z \prec y$. Thus $z \rightarrow y$, contradicting $y \rightarrow z$. $\square$

The natural extension of this argument replaces a k -cycle through a comparable edge by a $(k-1)$ -cycle which need not contain one, so it yields a contradiction only at $k = 3$. Rather than carry an unproved lemma into a computation of this size, the census builds the full relation $D(P)$,

which is what [10] defines, and computes the restricted spectrum alongside it as a separate set of counters. The comparison then becomes measured data. Adding the comparable edges costs nothing: they are read off the order relation and require no linear-extension counting.

4.2 The spectrum at order 14

n	any	3	4	5	6	7	8
9	5	5	0	0	0	0	0
10	153	148	6	0	0	0	0
11	5,815	5,740	101	0	0	0	0
12	218,097	216,573	2,885	5	21	0	0
13	9,348,400	9,318,881	102,127	471	363	1	0
14	478,632,938	477,954,774	5,419,981	33,473	10,423	61	30

Table 2: Classes carrying a simple majority cycle of each length. A class may carry cycles of several lengths, so the “any” column is not the row sum. Orders 9–13 reproduce [10]. No class at order 14 carries a cycle of length 9 or more.

The proportion of classes carrying a cycle continues to rise, as Gehrlein and Fishburn conjectured [15]: at order 14 it is 3.5767 per 10^4 , against 2.764 at order 13. De Loof et al. fitted $f(n) = a + bn + n \log n$ to their orders 9 through 13, obtaining $a = 5.2304$ and $b = -2.7572$, and expected it to approximate the incidence for n below about 50. Their fit gives $f(14) = 3.5764$, against the measured 3.5767. The residual is 0.00031, but one out-of-sample point cannot validate the functional form.

The counts fall steeply in length but not monotonically. At order 12 there are 21 classes with a 6-cycle and only 5 with a 5-cycle; at orders 13 and 14 the order reverses, though the 5-to-6 ratio stays far below the 3-to-4 ratio. Section 4.4 offers a structural reason why even lengths can be over-represented.

4.3 Cycles of length eight

Exactly 30 classes carry an 8-cycle. The census reports as its canonical witness the lexicographically least encoding,

14:0000000000000000000000000300050006000300050002000400f7035705a7,

in the encoding of Section 7, with $e(P) = 7,511,140$, $b(P) = 1/2$, width 8, height 3, and cycle spectrum $\{3, 4, 5, 6, 7, 8\}$.

Figure 3 draws a different member of the family,

14:00000000000000000000000003000500030005000300050002000401ff043f03c7,

chosen because its cycle spectrum is exactly $\{4, 8\}$ — it carries an 8-cycle and no odd cycle at all — which is the phenomenon of Section 4.4. It has three minimal elements, an eight-element antichain, and three maximal elements. For both posets the restricted digraph $D_{\text{inc}}(P)$ has the same spectrum as $D(P)$.

is 71.0% at length 3 (11,626 of 16,384), 49.8% at length 4 (8,162 of 16,384), 27.2% at length 5 (2,296 of 8,428), and 98.8% at length 6 (3,067 of 3,103). Exact balance is therefore common among cycle-carrying posets generally, and conspicuously not monotone in cycle length, so the value 100% at lengths 7 and 8 — over 28 and 13 witnesses — is suggestive rather than evidence of a universal mechanism. A transposing automorphism of an incomparable pair forces that pair to be exactly balanced [21, Prop. 2.3], and 35 of the 37 witnesses admit one. The remaining two have trivial automorphism group and are exactly balanced anyway.

Observation 10. The cyclic part of $D(P)$ is an antichain in P for every one of the 37 recorded posets carrying a cycle of length 7 or 8, but not in general: of the 44,013 distinct retained witnesses, 166 have a cyclic part containing a comparable pair. They are concentrated at length 6, where 118 of 3,103 do, against under 0.42% at every other length.

Thus Proposition 7 does not extend to the whole cyclic part: 166 retained witnesses contain a comparable pair among their cyclic vertices, although no directed 3-cycle uses a comparable edge. This is compatible with Observation 11: a comparable pair may lie in the cyclic part without any cycle using the edge between them.

For every order-14 class the census computes the cycle spectrum of $D(P)$ and of $D_{\text{inc}}(P)$ separately.

Observation 11. For every enumerated poset through order 14, the full and restricted majority digraphs have the same cycle spectrum.

This is an exhaustive computational conclusion, not merely a comparison of retained witnesses. For each length, the classes carrying that cycle in $D_{\text{inc}}(P)$ form a subset of those carrying it in $D(P)$, because the former is a subdigraph of the latter. The exhaustive counters give equal cardinalities for these two sets at every tested order and length, so the sets, and hence every class's spectrum, agree. Proposition 7 proves the length-3 case in general; the longer-cycle statement remains Question 14 below.

5 The Gold Partition Conjecture

5.1 Certificates

For an ordered incomparable pair (x, y) , Peczarski calls z a *slave* if either $z > x$ and $z \parallel y$, or $z < y$ and $z \parallel x$. The verifier uses the following sufficient conditions from Peczarski's Definitions 1–2 and Lemmas 1–2 [22, pp. 91–92].

Lemma 12 (Peczarski). *A non-chain P has the Gold Partition property if one of the following conditions holds.*

1. *There is an ordered incomparable pair (x, y) with at most one slave and*

$$2e(P + xy) \geq e(P).$$

2. *There are three distinct elements x, y, z such that*

$$e(P + xy + yz) \leq \max\{e(P + yx), e(P + zy)\} \leq \frac{e(P)}{2}.$$

The elements in condition (2) need not be pairwise incomparable. Peczarski’s separate cyclic-triple screen is subsumed by the exhaustive search for condition (2). In each case the proof supplies two comparisons fixed before their outcomes are known, as required by the conjecture.

The verifier also accepts a half-balanced incomparable pair (x, y) , meaning

$$2e(P + xy) = 2e(P + yx) = e(P).$$

Indeed, use this pair as the first comparison. Then $t_1 = t_0/2$, while any fixed second comparison leaves at most t_1 extensions; hence $t_0 \geq t_1 + t_2$. This direct argument is also recorded by Peczarski [24].

There is a useful inexpensive screen for condition (1). If both orientations of an incomparable pair have at most one slave, one of $e(P + xy)$ and $e(P + yx)$ is at least $e(P)/2$, so condition (1) holds without counting either quantity. The verifier applies this bilateral screen first. It then searches for a low-slave pair or a half-balanced pair, and finally for a triple satisfying condition (2). The division between the first two certificate counters depends on search order; their union does not, and only the union is reported.

Candidate pairs are tried in a simple structural order and candidate triples by decreasing right-hand bound in Lemma 12. In the production classifier two configurations need no new recurrence: $e(P + xy + yz) = 0$ when $z < x$, and when $x < z$ the positions of y relative to $x < z$ partition the extensions, giving $e(P) - e(P + yx) - e(P + zy)$.

5.2 The order-14 decision

	$n = 12$	$n = 13$	$n = 14$
isomorphism classes	1,104,891,746	33,823,827,452	1,338,193,159,771
chain	1	1	1
pair certificate	1,066,006,204	32,418,324,910	1,272,077,147,789
balanced-triple certificate	38,885,541	1,405,502,541	66,116,011,981
open	0	0	0

Table 3: Gold Partition certificate aggregates under a pair-first classification. “Pair certificate” means that a low-slave or half-balanced pair exists; “balanced-triple certificate” means that no pair certificate exists and a triple does. For each order the four rows partition the total.

The order-14 aggregate contains all 4,096 residues and agrees with the published number of unlabeled posets, first obtained by Heitzig and Reinhold and independently confirmed by Brinkmann and McKay [16, 6]. Exactly one class is a chain, the certificate counts partition the total, and the open count is zero, which is Theorem 5. The known total checks enumeration completeness; the partition and shard checks are internal consistency tests; the zero open count is the computational conclusion. The released aggregator was rerun on the preserved 4,096-shard archive, and the same executable reproduces the known class totals at orders 10 through 13 together with the certificate counts of an earlier run.

The share of classes needing the weaker balanced-triple certificate rises steadily: 2.65, 3.02, 3.52, 4.16, and 4.94 percent at orders 10 through 14. Peczarski observes that his Table I might suggest almost all posets fall to the low-slave condition and shows this is false: by a zero-one law the fraction of posets satisfying its hypothesis tends to 0 [22, p. 94]. The five measured orders are consistent with that result, but the reported pair category also includes half-balanced pairs, so it

does not by itself imply that the balanced-triple share must rise. How that share behaves, and whether the certificate scheme stays practical at orders we cannot reach, the data does not say.

The order-14 calculation took 1,694 user core-hours over a 19-hour shard span on AMD EPYC 7B13 processors; the census took 5,103 user core-hours over a 53-hour shard span on the same hardware, the difference being that a census cannot stop at a certificate. In a controlled ablation on the production platform, removing only the bilateral early exit increased median user time by factors of 4.25, 4.81, 5.57, and 8.55 at orders 10 through 13, respectively. The first figure is a complete pass; the others aggregate the same three fixed modulus shards over alternating repetitions. The baseline binary is the production binary, and the total, pair-certificate union, balanced-triple count, and open count agreed in every comparison. This measures the screening step within the present implementation, not a speedup over the programs of Peczarski or De Loof et al.

6 Where the extremal posets sit

Sections 3 and 4 produce two families of extremal objects; Section 5 partitions every class by the certificate that resolves it. Neither program records what the other measures, so we classified the recorded extremal witnesses with an independent implementation of Lemma 12 (Section 7).

These are the witness sets the census retains — one per distinct extremal value, one per cycle length per shard — not samples drawn from the classes, and they cannot estimate class-wide rates. How much that matters is visible directly: of the 42 one-per-value low-tail witnesses 21 have no pair certificate, while of the 180 distinct low-tail witnesses retained across all shards 64 also have no pair certificate. The same property, two retention rules, 50% against 35.6%. For scale only, 4.94% of all order-14 classes need the balanced triple (Table 3); that exhaustive class rate is not commensurable with either witness fraction.

The two exhaustive runs do give one population-level intersection. The twelve low-tail values first attained at order 14 contain all 17 non-ordinal-sum classes in the recorded tail: the 14 classes at the nine lower values require a balanced triple, while the three classes at the three higher values have a bilateral pair certificate. Each value is one complete duality orbit, so classifying its representative and dual exhausts the attainment set.

The following statements concern named retained witnesses and do not infer population rates. $L_{14,1,9}$, the worst-balanced non-sum poset of Theorem 1, has *no* pair certificate and is resolved only by a balanced triple. Every one of the 37 recorded posets carrying a cycle of length 7 or 8 has balance $1/2$ and hence a half-balanced pair certificate. In addition, all 37 are caught by the bilateral screen, which needs both orientations to have at most one slave; this stronger fact is observed rather than forced. And the families separate by width: every retained low-tail witness has width 2 or 3 and every equality representative width exactly 2, against 7 to 10 for the long-cycle witnesses. Every equality-family class also has a bilateral certificate: in a $T = \{a < b, c\}$ summand the ordered pair (a, c) has the sole slave b , while (c, a) has none, and every other ordinal summand is comparable with both elements.

Observation 13. Every recorded order-14 poset carrying a majority cycle has balance constant at least $2869/5810 = 0.4938\dots$, while the low tail is bounded above by $9/25$.

The disjointness alone is weaker than the observed balance separation, and no overlap probability can be inferred because the witnesses are retained deterministically rather than sampled from the classes. What Observation 13 adds is that the separation is not marginal — the two families sit at opposite ends of the balance range, one pressed against $1/3$ and the other against

1/2. Settling any of this over all classes would need a pass recording both statistics together; the existing shards store marginals and selected witnesses, so it cannot be done retrospectively.

7 Verification

A witness is written `n:xxxx...`, the order followed by n four-digit hexadecimal upper-set masks: bit y of mask x is set exactly when $x < y$ in the strict transitive closure. The decoder validates irreflexivity, antisymmetry, and transitive closure before use, so a malformed or non-transitive string is rejected rather than silently interpreted.

We separate three kinds of check, because they carry very different weight.

External ground truth. Both runs' class totals agree exactly with the published number of unlabeled posets on 14 points [16, 6]. A single missing, duplicated, or truncated shard would shift a total by roughly 10^8 , so digit-exact agreement is a strong check on enumeration completeness and on the disjointness of the residue partition. The census cycle counts at orders 9 through 13 agree exactly with [10], and the ladder values with Table 1 of [25].

Two further external checks are available and both pass. First, Peczarski's Table I [22] supplies an external regression for the certificate logic. It gives, for $4 \leq n \leq 11$, the number of non-chain ordinal-indecomposable classes with dual pairs identified together with the number resolved by each criterion. Rebuilding that universe and classifying it with an independent implementation of Lemma 12 reproduces both his class counts and his pair/triple split exactly at every order our independent generator reaches, which is $4 \leq n \leq 8$.

At orders 4 through 8 he records respectively 6, 21, 111, 725, and 6,474 classes. Their pair-certificate counts are 6, 19, 105, 704, and 6,322, while the balanced-triple counts are 0, 2, 6, 21, and 152; the independent classification returns every number exactly.

This is the only check that tests the certificate conditions against an external source rather than against the enumeration, and it is the check the order-14 decision most needs, since $\text{open} = 0$ is otherwise self-reported.

Second, the census's count of ordinal-indecomposable classes is externally determined. Ordinal decomposition into indecomposable summands is unique, so if $A(x) = \sum_n a(n)x^n$ with $a(0) = 1$ and $a(n)$ the number of unlabeled n -element posets, and $B(x)$ is the corresponding series for the indecomposables, then $A = 1/(1 - B)$. Inverting against the published totals [16, 6] predicts 1,269,310,589,336 indecomposable classes at order 14 and 31,559,446,774 at order 13; the census counts exactly those. This validates the ordinal-sum decomposition on which Theorem 1 depends and which nothing else in the pipeline tests.

The order-14 extremal data itself has, by construction, nothing external to be checked against.

Independent recomputation. Every extremal witness is recomputed from its witness string alone by a program sharing no algorithm with the census: where the census multiplies prefix and suffix counts over the ideal lattice as in (2), the checker counts, for each ordered pair separately, the linear extensions with that pair's relation adjoined, by a plain forward recurrence. At orders 5 through 9 the census is compared on every class against a brute-force reference that enumerates linear extensions directly.

For the Gold Partition pass, a Python program generates every unlabeled poset through order 7 and applies the certificate definitions to explicitly enumerated extensions, and a second implementation using a plain subset recurrence is compared with the released classifier on every

poset at orders 8 and 9, including non-topological labelings. Neither shares extension-counting or certificate code with the classifier. No independent per-poset check is made above order 9.

Internal consistency. For each run: the residues are disjoint and covering; every shard carries exactly one summary and one parameter line; every shard records the same binary hash; the census payloads match the SHA-256 sealed into their metadata before publication; the census equality, tail, and cycle-budget counters report no overflow; the certificate counters partition the total; and exactly one class is a chain. These checks would not detect a classifier that is consistently wrong, which is why they are listed last.

Provenance. Each production binary hash is conditional on the recorded native build, GCC 13.3.0 on AMD EPYC 7B13 with `-march=native`: independent builds on two hosts of that model are byte-identical, which does not generalise to other silicon. A portable build is provided and produces identical output, as do GCC 13, 14, and 16 at orders 10 and 11.

8 Open questions

Question 14. Do $D(P)$ and $D_{\text{inc}}(P)$ have the same cycle spectrum for every finite poset? Proposition 7 settles length 3; Observation 11 is the evidence for the rest.

Question 15. Do posets whose majority digraph has bipartite cyclic part admit a characterisation? Observation 8 gives a sufficient condition for the absence of odd Condorcet cycles, and only a sufficient one: the converse fails already at order 14, where retained witnesses with cycle spectrum exactly $\{4\}$ have a cyclic part that is not bipartite. An undirected odd cycle need not be consistently oriented, so it need not carry a directed one.

Question 16. Does Peczarski’s broken-rung conjecture hold for all $n \geq 9$?

Question 17. Is the family of Proposition 3 the complete equality locus of the $1/3$ – $2/3$ Conjecture at every order?

De Loof et al. also study cycle-free cut levels, which require repeated cycle detection at many thresholds. We did not compute them; it is the natural follow-up to this census.

9 Reproducibility

The manuscript, source, tests, and compact aggregate data are available at

<https://github.com/agupta/gold-partition-conjecture>.

The compact aggregate reports for both runs are included in the repository under `data/`. The complete order-14 shard archives, their checksum manifests, and the build records are archived on Zenodo. The concept DOI 10.5281/zenodo.21576029 resolves to the latest version; readers checking the data behind *this* paper should use the version DOI recorded in `data/census-n14.txt`, since the concept DOI will move when a later version is deposited. The shard archive retains witness strings for every extremum and tail value, every equality representative, and one witness per cycle length per shard. The compact aggregate enriches its global extrema with canonical and Hasse `digraph6` encodings. Witness-specific claims can be recomputed from those

strings; global counts, completeness, and the equality classification are checked by the aggregator over the complete sealed shard set. The retained-witness statistics are recomputed by `scripts/analyze_witness_archive.py`. The figures are generated from fixed witness strings by `scripts/make_census_figures.py`, which re-verifies the asserted properties before drawing them. Each archive includes the source files used to build the corresponding production binary.

The calculations used `genposetg` 1.1 from nauty 2.9.1 and native GCC 13.3.0 builds on AMD EPYC 7B13 processors. The repository build is portable by default. Tests of the shard protocol and aggregation rules are included with the source, and the `make check` target runs the independent and differential small-order checks, aggregation tests, and an end-to-end sharding test.

An earlier version of this paper reported the Gold Partition result alone, without the census of Sections 3–4.

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